TEXORA · LECTURE SHEET

Theory of Machine

How machines move: links, pairs and degrees of freedom, the four-bar family that drives needles, beat-up and shedding motions, velocity ratios of belts and gears, simple and compound gear trains — and the flywheel and governor that smooth and regulate it all.

ME-3210 · SOLID MECHANICS AND MACHINE DESIGN · TOPIC 6 · LEVEL 3 TERM 2

1. Links, pairs and degrees of freedom

A **link** is a rigid body connecting joints; a **kinematic pair** is the joint itself — **lower pairs** (surface contact: turning/revolute, sliding/prismatic — pins, sliders) and **higher pairs** (line or point contact: cams, gear teeth, rolling wheels). A chain of links is a **mechanism** when one link is fixed and motion is constrained — one input determines every position. Mobility is counted by **Gruebler's criterion** for planar motion:

$$\mathrm{DOF} = 3(n - 1) - 2j - h$$

with n links (including ground), j lower pairs, h higher pairs. A four-bar (n = 4, j = 4): DOF = 3 × 3 − 2 × 4 = **1** — one motor drives it completely. DOF 0 is a structure; DOF 2 needs two inputs (a differential, a five-bar robot leg). Cam-and-follower and gear pairs appear throughout the mill: shedding cams, dobby drives, winding drums, take-up gearing.

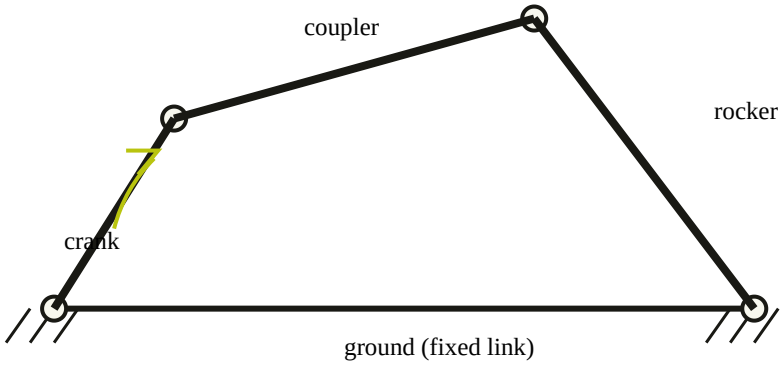


Figure 1 — The four-bar linkage: crank (full rotation), coupler, rocker (oscillation) on a fixed ground link — one degree of freedom, the ancestor of half the motions in a weaving shed.

2. The four-bar family and inversions

Grashof's law sorts behaviour: if shortest + longest ≤ sum of the other two, the shortest link can rotate fully. Fixing different links of the same chain gives the family: crank-rocker (rotary → oscillation: windscreen wipers, loom sley drive via crank and connecting rod), double crank (rotary → rotary, uneven), double rocker, and the **slider-crank** (one revolute replaced by a sliding pair: engines, compressors, needle bars) — crank rotation becomes slider reciprocation with a quick-return flavour when geometry offsets. The oscillating rocker output, timed by crank angle, is precisely what beat-up and shedding motions order from a mechanism designer.

3. Belt, chain and gear velocity ratios

For friction and gear drives the surface speeds at contact match (belts approximately, gears exactly):

$$N_{driven}/N_{driver} = D_{driver}/D_{driven} \text{ (belts, diameters)} = T_{driver}/T_{driven} \text{ (gears, teeth)}$$

WORKED EXAMPLES — STEPS AND COMPOUNDS

Simple gear pair: a 20-tooth pinion at 1000 rpm drives a 60-tooth wheel: N = 1000 × 20/60 = **333 rpm**, direction reversed, torque (ideally) tripled.

Compound train: A (20 T) meshes B (60 T); C (25 T) is fixed on B's shaft and meshes D (75 T). Overall ratio = (60/20) × (75/25) = 3 × 3 = 9, so 1000 rpm in gives **111 rpm** out, same direction as input (two external meshes). Compounding is how a compact gearbox buys a 9:1 reduction that a single pair could not house.

Belt with slip: a 150 mm pulley at 1440 rpm drives a 450 mm pulley: ideal driven speed = 1440 × 150/450 = 480 rpm; with 2% slip, **470 rpm** — the small, real tax friction drives pay, and the reason timing-critical drives use toothed belts or gears instead.

An **idler** gear changes direction and reach but not ratio (its teeth cancel). **Epicyclic (planetary) trains** — sun, planets, ring, carrier — pack large ratios and coaxial shafts into small volume and appear in automatic gearboxes and winders; their analysis fixes one member and applies relative-velocity (tabular) bookkeeping.

4. Flywheel and governor — smoothing and regulating

A **flywheel** stores kinetic energy ($E = \tfrac{1}{2}I\omega^2$) to absorb torque fluctuations *within* a cycle: the punching machine and the engine's idle smoothness are its work — sized from the coefficient of fluctuation of speed the process tolerates. A **governor** works *between* cycles: it senses mean speed and moves the fuel/steam valve to hold it as load changes (centrifugal Watt/Porter governors on older engines,

electronic governors on modern DG sets). Confusing the two is a classic error: no flywheel can correct a sustained overload, and no governor can smooth a single-cycle torque spike.

5. Key points

- Gruebler: DOF = 3(n − 1) − 2j − h; the four-bar has exactly 1 — one input, fully determined motion.
- Grashof decides who rotates fully; slider-crank is the four-bar's reciprocating inversion (engines, needle bars, sley drives).
- Velocity ratio = inverse teeth/diameter ratio; compound trains multiply stage ratios ($3 \times 3 = 9 \rightarrow 111$ rpm from 1000).
- Belts slip (2% typical tax); gears do not — choose the drive by how much timing matters.
- Flywheel smooths within the cycle; governor regulates mean speed against load changes.

6. Practice

Answer in your own words; brief model answers follow.

1. Find the DOF of a five-bar linkage (5 links including ground, 5 revolute joints).
2. A slider-crank has crank 50 mm and connecting rod 200 mm. Find the stroke, and the crank angle positions (measured from inner dead centre) at which the slider is at mid-stroke — or state what makes the question trickier than it looks.
3. A gear train: 18 T pinion at 1200 rpm drives 54 T; on the same shaft a 20 T pinion drives 80 T. Find the output speed.
4. An idler is inserted between a 24 T driver and a 48 T driven gear, driver at 900 rpm. Find the driven speed and the direction relative to the driver.
5. A flywheel must absorb 2 kJ per cycle while its speed varies between 198 and 202 rpm. Estimate the required moment of inertia (use mean speed 200 rpm).
6. In a weaving shed, a sley must dwell near front centre while the shuttle crosses. Which mechanism features produce dwell, and why is a plain crank-and-rod only an approximation to it?

Answer key

1. DOF = $3 \times 4 - 2 \times 5 = 2$ — two independent inputs are required for determined motion.
2. Stroke = 2 × crank = 100 mm. Mid-stroke is *not* at 90°/270° exactly: obliquity of the rod shifts the slider position $x = r\cos\theta + \sqrt{(l^2 - r^2\sin^2\theta)}$; with $l/r = 4$ the offset from 90° is small (≈ 0.9 mm of travel) but real — the slider reaches mid-travel slightly after the quarter turn, which is the seed of quick-return asymmetry.
3. Ratio = $(54/18) \times (80/20) = 3 \times 4 = 12 \rightarrow$ output = 1200/12 = 100 rpm.

4. Idler cancels: ratio = 48/24 = 2 → 450 rpm, and the driven gear rotates in the *same* sense as the driver (two external meshes). Idlers set direction and span distance, not ratio.
5. $\omega_{mean} = 2\pi \times 200/60 = 20.94$ rad/s. $\Delta E = I\omega^2 C_s$ with $C_s = (202 - 198)/200 = 0.02$: $\Delta E \approx I \times (20.94)^2 \times 0.02 = 8.77\ I$. $I = 2000/8.77 \approx 228$ kg·m² — a serious wheel, which is why fluctuation allowances, not wishes, size flywheels.
6. Dwell comes from geometry that parks the output: six-bar dwell linkages (a coupler point tracing a near-straight/circular arc), cam profiles with a constant-radius sector (the shedding tappet's dwell — Section TE-2103), or Geneva-type intermittent drives. A plain crank-rocker's rocker only slows near its extremes (sinusoidal-ish motion), never truly stops, so the shuttle window is approximate and speed-limited.

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